Gavin Gleinig

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SUMMARY Software engineer with graduate research in computer vision and AI security. Owns systems end to end: scoping them with the people who use them, then building the pipelines, web apps, and ML behind them with agentic coding tools. U.S. citizen, clearance-eligible.

EDUCATION

Texas A&M University, College Station, Texas
Master of Computer Science, May 2026
4.00 GPA, Coursework in AI / ML
Bachelor of Science in Computer Science, May 2025
3.98 GPA, Minor in Mathematics

TECHNICAL SKILLS

Languages: Python, SQL, C++, Julia
Frameworks: PyTorch, Pandas, NumPy, scikit-learn, Hugging Face
AI / ML: MCP, Tool-Use Agents, Agentic Coding Tools, LLM Evals, RAG, Computer Vision
Tools & Certifications: Git, Docker, Linux, Cloudflare, PostgreSQL, CompTIA Security+

WORK EXPERIENCE

Software Engineer May 2026 – Present
Independent Consultant: Gloxco, ETEC, Gtek, Tarpon Specialty Services, Mr Gatti's Pizza
- Unified 10 POS, payroll and arcade feeds into one warehouse and daily owner scorecard
- Piloted AI that turns handwritten field test sheets into database records, no retyping
- Launched bilingual scan-to-win reward app for 540-worker crew; live since Sep 2026
- Built in-house support inbox to retire paid helpdesk; AI triages mail and drafts replies

E-Commerce Operations Intern Summers 2022 – 2025
Gloxco, Alice, Texas
- Optimized picking and shipping workflows for 1,400-SKU warehouse; trained staff on them
- Wrote small Python scripts to cut manual entry of shipment tracking and inventory data

IT Project Intern May 2025 – July 2025
City of Portland, Portland, Texas
- Deployed IDS across 5 municipal networks for real-time threat monitoring
- Investigated and remediated security incidents including phishing-driven malware

PROJECTS

Adversarial Robustness Testing Framework for Vision Models (AI Security Research)
- Built validation framework, testing vision model robustness against adversarial attacks
- Implemented differentiable image transforms with end-to-end gradient-based optimization
- Achieved 72% attack success at 0.129 LPIPS, matching SOTA diffusion attacks

Sparse-View 3D Reconstruction with Generative Repair
- Engineered 3D reconstruction pipeline using Gaussian Splatting and diffusion repair
- Fine-tuned LoRA across 1.4B-parameter Diffusion U-Net, ControlNet, and CLIP backbone
- Improved Peak Signal-to-Noise Ratio (PSNR) by 41% over baseline sparse reconstruction

Subquadratic MST Approximation for Large-Scale Clustering
- Implemented subquadratic Metric Forest Completion clustering framework in Julia
- Matched exact-MST clustering quality (ARI/NMI) on 20k-point synthetic Gaussian data
- Outperformed exact MST on noisy genomic data at subquadratic cost

ACTIVITIES

Corps of Cadets, Fightin' Texas Aggie Band Aug 2021 – May 2025
- Training Officer: led 10-person team responsible for wellness of a 60+ member unit
- Scholastics Sergeant: mentored cadets to 3.4 unit-wide GPA & Most Outstanding Outfit

OR Simpson Honor Society Aug 2022 – May 2025
Honor Society serving Cadets through tutoring and mentorship